Justin Manoj

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Education

University of Massachusetts Amherst — Amherst, MA
Bachelor of Science in Computer Science and Mathematics – GPA: 3.8/4.0 — *Expected Graduation: May 2027*

Experience

Founding Engineer — Apr. 2025 – Present
ChitChat Workplace – Venture-Backed Startup — *Boston, MA*

- Launched *ChitChat* and the core platform MVP that centralizes messaging, email, and third-party apps, successfully onboarding initial enterprise customers and reducing context-switching by **10+** hours/month.
- Architected the core messaging engine using Go (Golang) and WebSockets, supporting **50k+** concurrent users.
- Designed a distributed event pipeline with Apache Kafka and Redis Pub/Sub to handle **2M+** daily events (messages, reactions, typing indicators), decoupling real-time delivery from persistent storage.
- Shipped "Smart Context" AI features using LangChain and Vector Embeddings (Pinecone), enabling semantic search and auto-drafting that reduced repetitive internal queries by **40%**.

Software Engineer Intern — Mar. 2025 – Sept. 2025
GBCS Group — *Alberta, CA*

- Re-engineered the backend data layer by migrating **8** REST endpoints to GraphQL, eliminating payload overfetching by **40%** and simplifying frontend state management.
- Optimized full-stack performance by implementing PostgreSQL composite indexes and React Query caching strategies, slicing average page load times by **65%** for core dashboards.
- Shipped reusable TypeScript and Tailwind components and strengthened code quality by implementing linting rules, Jest utilities, and component documentation.

Undergraduate Research Assistant — Dec. 2024 – Feb. 2025
University of Massachusetts Amherst - Autonomous Learning Lab — *Amherst, MA*

- Built and ran automated stress-tests for **7** multi-agent AI frameworks (like AutoGen and MetaGPT) using Python, identifying **14** critical bug types that cause AI agents to fail in production.
- Processed and labeled **1,600+** execution logs using OpenAI API and Pandas, creating a standardized "crash dataset" used to benchmark model reliability scores.
- Developed a failure-detection pipeline that automatically flags errors (like infinite loops or hallucinations) in agent conversations, reducing manual review time by **70%** for research purposes.

Projects

UMass Dining Recommendation Engine | *Java, Spring Boot, React, PostgreSQL* — June 2025

- Launched a personalized nutrition platform for **30,000+** students, aggregating daily menus from **4** dining halls to generate AI-driven meal plans; pending integration into the official UMass Dining App.
- Built backend using Spring Boot and PostgreSQL, implementing Redis caching to serve real-time menu queries.
- Engineered a robust data ingestion pipeline with Python schedulers that scrapes, cleans, and normalizes **100+** daily menu items, ensuring **99.9%** data accuracy for dietary filters.

Multi-User Payment Hub | *Typescript, Next.js, PostgreSQL* — February 2025

- Built a secure group payment platform that holds funds in escrow until approval and tracks user "Reliability Scores" in PostgreSQL, enabling informed split decisions beyond simple payment tracking.
- Implemented logic for "payment staking," where users deposit collateral that is automatically forfeited if they miss deadlines, reducing dispute rates and ensuring on-time settlements via Stripe.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, Go (Golang), SQL, HTML/CSS
Frameworks & Libraries: React Native, React, Node.js, Flask, Django, Express.js, PyTorch, Scikit-learn
Cloud & DevOps: AWS (Lambda, EC2, S3), Google Cloud Platform, Docker, Kafka, Redis, Kubernetes, CI/CD, Git
Tools & Concepts: RESTful APIs, GraphQL, Microservices, Agile Methodologies, Test-Driven Development